Yufan Du

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Education

Peking University **Sep. 2021 - Present**
Bachelor of Science in Applied Physics *and* Computer Science Dual Major *Beijing, China*

Relevant Scores

Overall GPA:	3.850/4.000 (Rank: 3rd out of 64 students)
TOEFL:	108 (R27 + L30 + S23 + W28)
GRE:	324 (Quantitative170 + Verbal154), 4.0 (Analytical Writing)

Technical Skills

Knowledgeable with: C/C++, Python
Familiar with: Python ML libraries (PyTorch, NumPy), Verilog, FPGA, Embedded systems, Linux systems, MATLAB, Tcl, LaTeX, SQL/Database, JavaScript/HTML
EDA tools: Familiar with Cadence Innovus, Cadence Virtuoso

Publications

- **Yufan Du**, Zizheng Guo, Yibo Lin, Runsheng Wang and Ru Huang, "Fusion of Global Placement and Gate Sizing with Differentiable Optimization," *2024 International Conference on Computer-Aided Design (ICCAD24).* (**Best Paper Award Nomination**).
- Zizheng Guo, Zuodong Zhang, Wuxi Li, Tsung-Wei Huang, Xizhe Shi, **Yufan Du**, Yibo Lin, Runsheng Wang and Ru Huang, "HeteroExcept: A CPU-GPU Heterogeneous Algorithm to Accelerate Exception-aware Static Timing Analysis," *2024 International Conference on Computer-Aided Design (ICCAD24).*
- **Yufan Du**†, Zizheng Guo†, Xun Jiang, Zhuomin Chai, Yuxiang Zhao, Yibo Lin, Runsheng Wang and Ru Huang, "PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning," *2024 Design Automation Conference (DAC24).*

Project Experiences

FPGA-Based Gesture-Controlled Snake Game **Mar. 2023 - July. 2023**
- Advisor: Professor Xiaohui Duan.
- Deployed *computer vision algorithms* (from Canny edge detection, circular hough transformation, to pattern recognization) and the classic Snake game on a resource-limited FPGA platform.
- Developed and implemented a *real-time control mechanism* with a camera detecting gestures.
- Provided insights for *IoT edge device.*

AI-Assisted Schedule Manager **Jan. 2023 - Apr. 2023**
- Advisor: Professor Wei Guo.
- Developed a *cross-platform application* integrated with commercial AI API for users' agenda scheduling.
- Better user experience through conversational interfaces for management and reminders.

Research Experiences

CUDA-Accelerated Gate Sizing **Jun. 2024 - Sept. 2024**

- Advisor: Professor David Z. Pan.
- Served as the first author.
- For ICCAD 2024 contest problem C.
- CUDA programming to accelerate the gate sizing process with gradient descent method.

The Integration of Gate Sizing and Global Placement **Dec. 2023 - May. 2024**

- Advisor: Professor Yibo Lin.
- Served as the first author.
- A novel fusion of global placement and gate sizing with differentiable optimization for broader optimization space.
- Demonstrated a substantial improvement in runtime efficiency and PPA metrics compared to traditional methods.
- *Accepted by ICCAD 2024 (Best paper candidate of track).*

Cross-Stage Power Prediction for Integrated Circuits **Jun. 2023. - Nov. 2023.**

- Advisor: Professor Yibo Lin.
- Served as the first author.
- ML prediction framework that integrates cross-stage circuit-transformation-aware learning.
- Achieved a significant reduction in error rates and computation time compared to industry-leading commercial tools.
- *Accepted by DAC 2024.*